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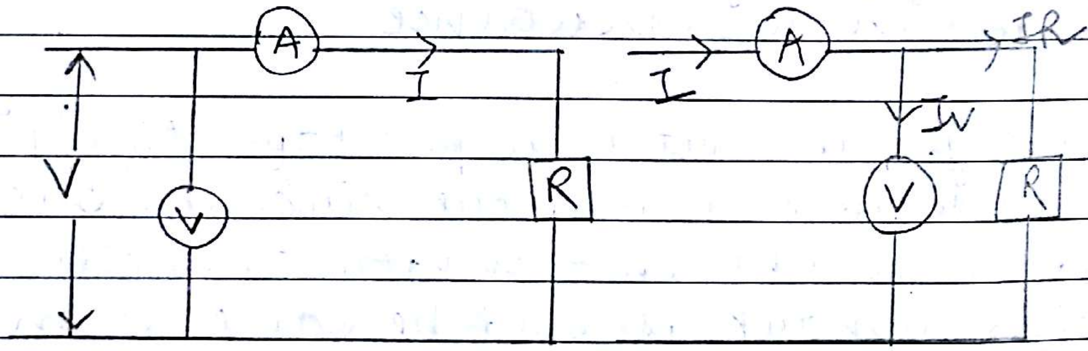
Resistances

Classification : (i) Low Resistances ($\leq 1\Omega$)
(ii) Medium Resistances ($1 \leq R < 0.1 M\Omega$)
(iii) High Resistances ($0.1 M\Omega$ and above)

(I) measurement of medium resistances.

(i) AMMETER - VOLTMETER METHOD



(a) (b)

(a) $R_{m1} = \dfrac{V}{I} = \dfrac{V_R + V_a}{I} = R + R_a$

$\Rightarrow R = R_{m1} - R_a$

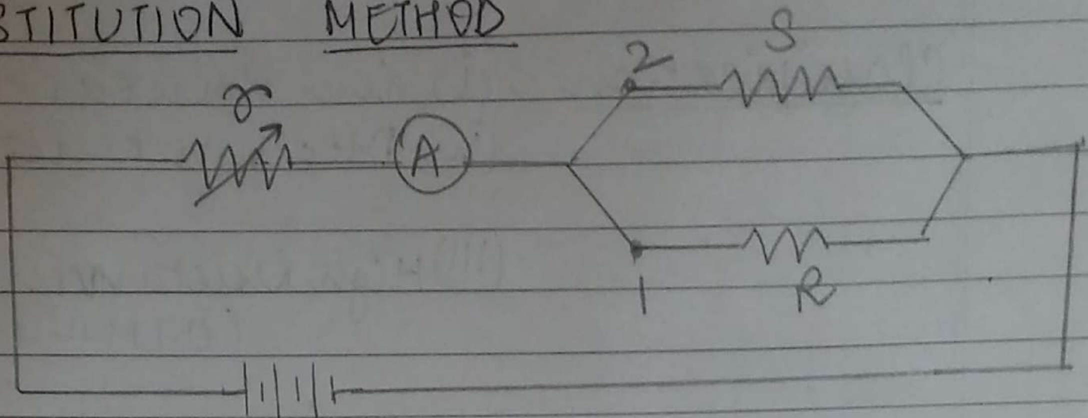
Relative error $(\varepsilon_r) = \dfrac{R_{m1} - R}{R} = \dfrac{R_a}{R}$

(b) $R_{m2} = \dfrac{V}{I} = \dfrac{V}{I_R + I_V} = \dfrac{V}{\dfrac{V}{R} + \dfrac{V}{R_V}}$

$\therefore R = \dfrac{R_{m2} R_V}{R_V - R_{m2}} = R_{m2}\left[\dfrac{1}{1 - R_{m2}/R_V}\right]$

$\varepsilon_r = \dfrac{R_{m2} - R}{R} = -\dfrac{R_{m2}}{R_V} \simeq \dfrac{-R}{R_V}$

(ii) SUBSTITUTION METHOD



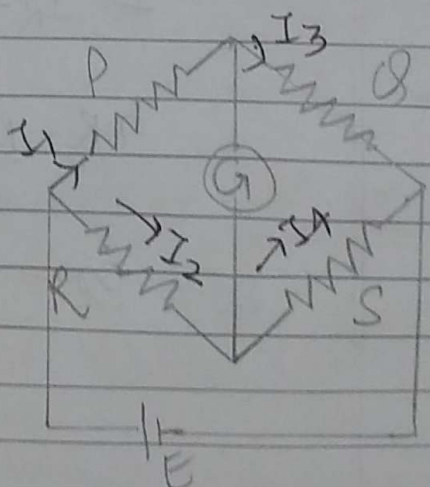
R = unknown Resistance
r = regulating resistance
S = standard resistance.

Firstly the switch is put to position '1' and R is connected in the circuit. The deflection obtained in the ammeter at the chosen mark should be same when the switch is put to position 2.

→ So the value of unknown resistance R is equal to S.

→ This is more accurate as compared to ammeter voltmeter method.

(III) WHEATSTONE BRIDGE



PQ → ratio arms
S → standard arm

$$R = \frac{PS}{Q}$$

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sensitivity of WSB

deflection of $G(\theta) = S_v e$ → voltage sensitivity

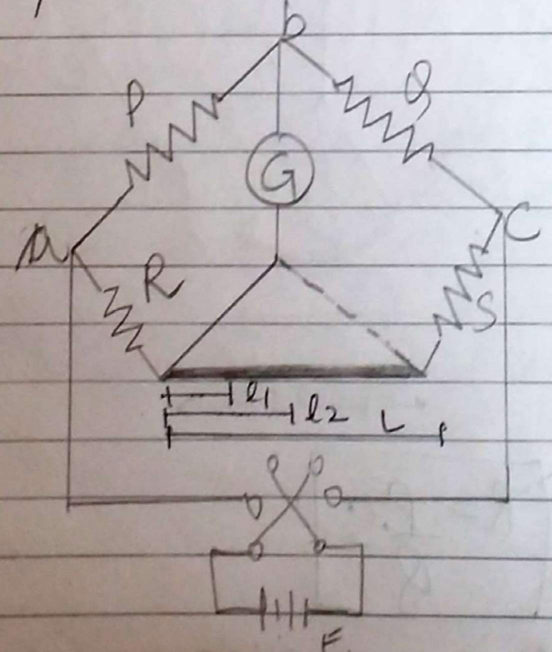
$$\boxed{\theta = S_v \cdot \frac{E\,S(\Delta R)}{(R+S)^2}}$$

where ΔR is change in Resistance

$$\boxed{S_B = \frac{\theta}{\Delta R/R} = \frac{S_v\, E\, S\, R}{(R+S)^2}}$$

S_B = Bridge sensitivity

(iv) CAREY FOSTER SLIDE WIRE BRIDGE



* The difference between S and R is obtained by Resistance per length of slide wire with the difference $(l_1 - l_2)$ between two slide wires at balance.

$$\boxed{R = \frac{S(l_1' - l_2') - S(l_1 - l_2)}{(l_1' - l_2' - l_1 + l_2)}}$$

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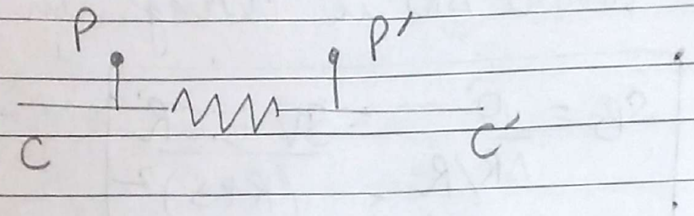
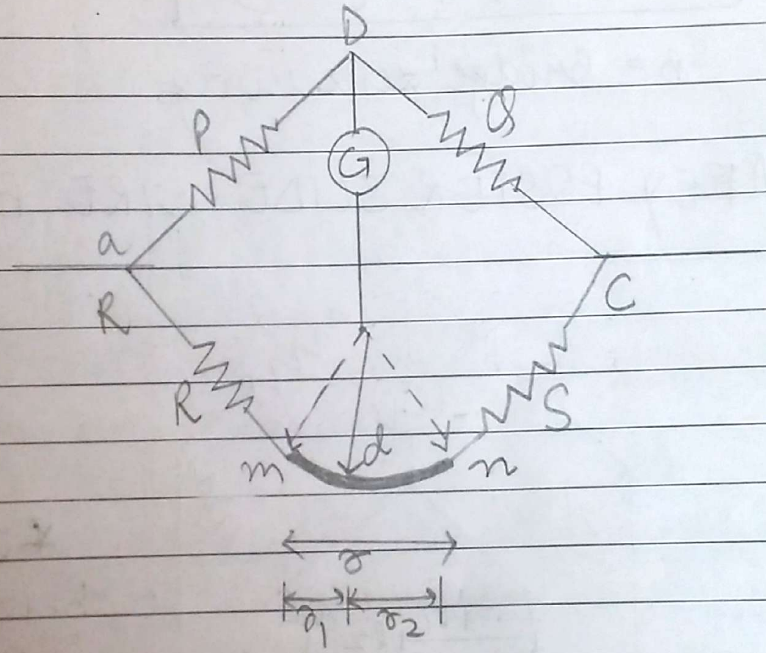
it is obtained by shunting either S or R by a known Resistance and again determining diff. in length ($l_1 - l_2$)

MEASUREMENT OF LOW RESISTANCES.





$$\boxed{R = \frac{P}{Q} \cdot S}$$

Kelvins bridge is a modification of Wheatstone bridge and provids increased accuracy in measurement of Low Resistance.

The above mentioned process is not practically feasible. Hence we use the next method.

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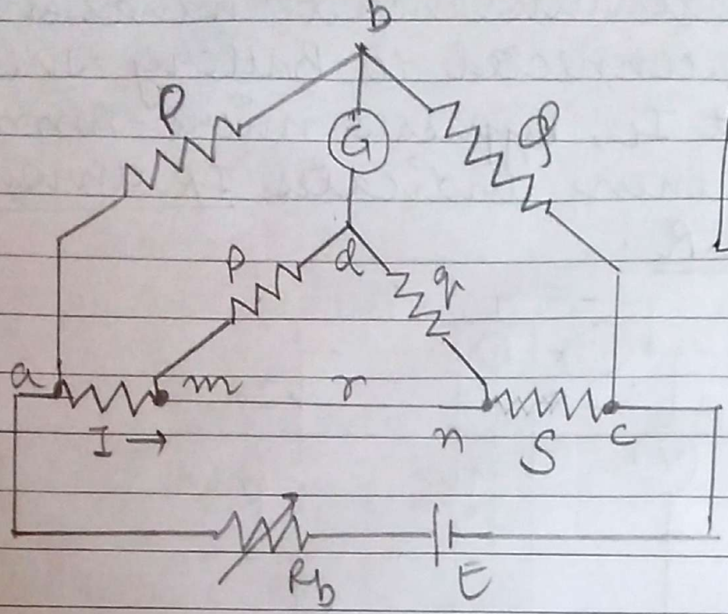
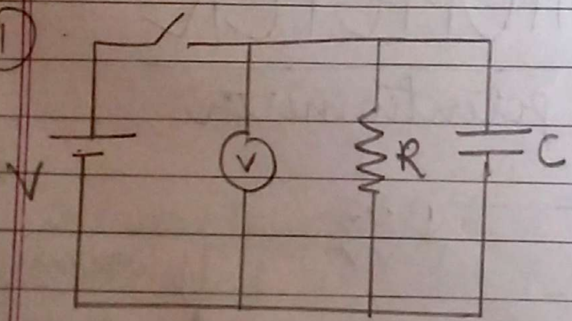
- KELVIN DOUBLE BRIDGE METHOD



$$\boxed{R = \frac{P}{Q} S}$$

MEASUREMENT OF HIGH RESISTANCE

(1)



C is precharged & then allowed to discharge through R.

$$\boxed{V_C = V e^{-t/CR}}$$

loss of charge method. R_1 = leakage resistance of capacitor.

$$R' = \frac{0.4343\, t}{C \log_{10} V/v}$$

R' is the resistance of R_1 and R in parallel

$$\boxed{R' = \frac{R R_1}{R + R_1}}$$

putting value of R_1, we can get value of R.

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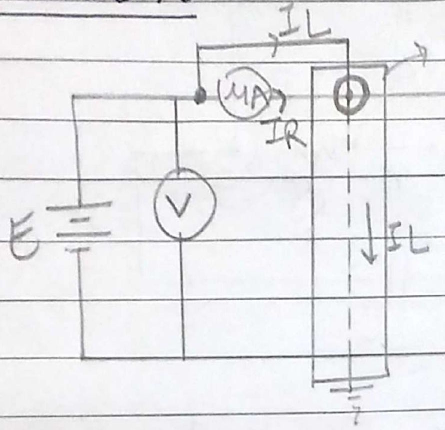
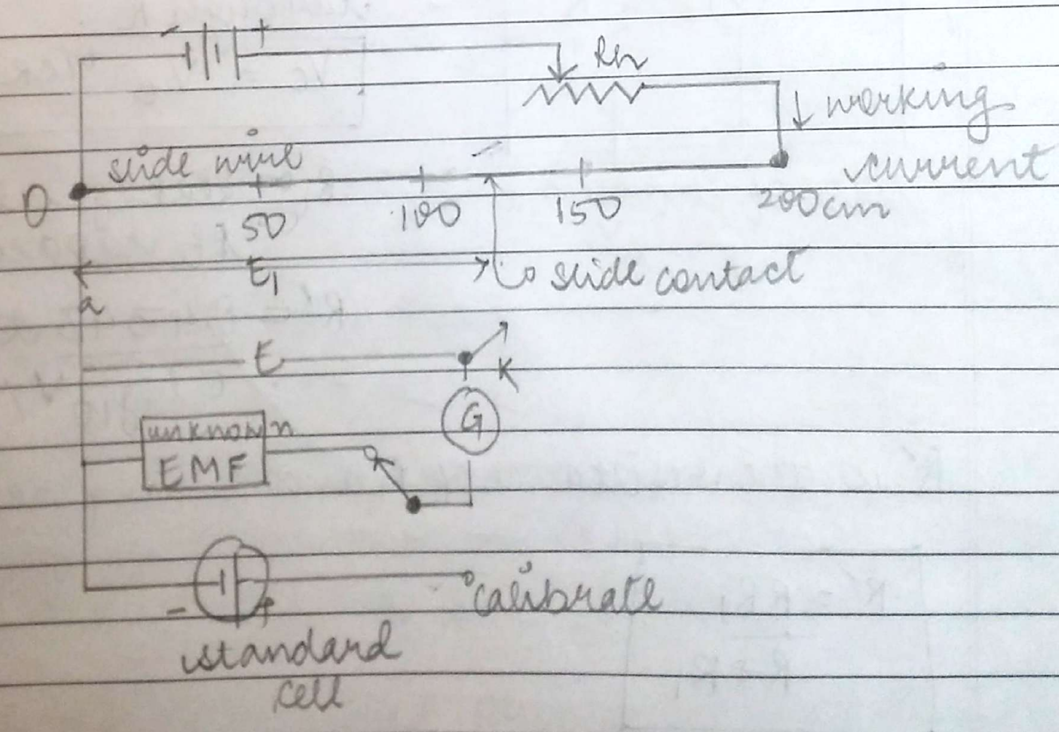
(2) <u>use of guard circuit</u> - The guard terminal surrounds resistance terminal entirely and is connected to Battery. The leakage current I_L, bypasses micro-Ammeter which then indicates <u>I_R through resistor R</u>.



POTENTIOMETER

Basic slide wire potentiometer



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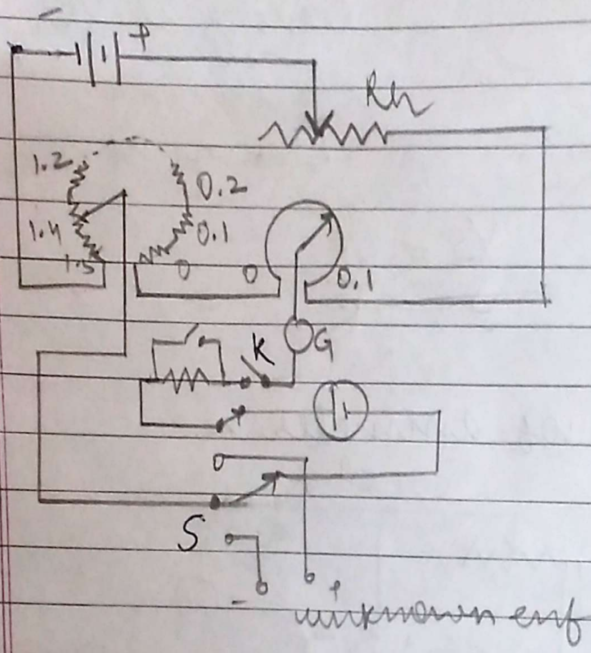
The unknown voltage E is equal to the voltage drop E_1 across ac portion of wire which is given by zero galvanometer deflection.

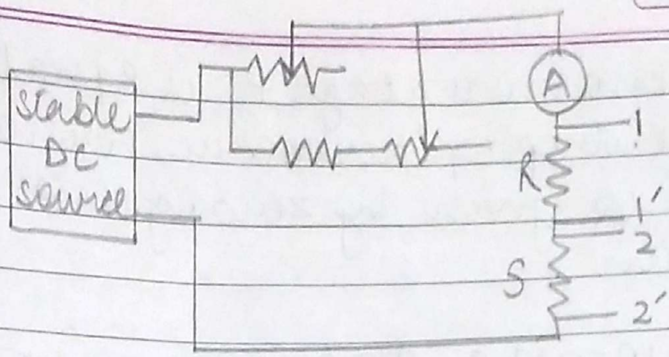
CROMPTONS POTENTIOMETER



The combination of dial resistors and slidewire is set to standard cell voltage. These are modern type potentiometers ~~use~~ which use calibrated dial resistors and small circular wire of one or more turns.

APPLICATION OF DC POTENTIOMETER

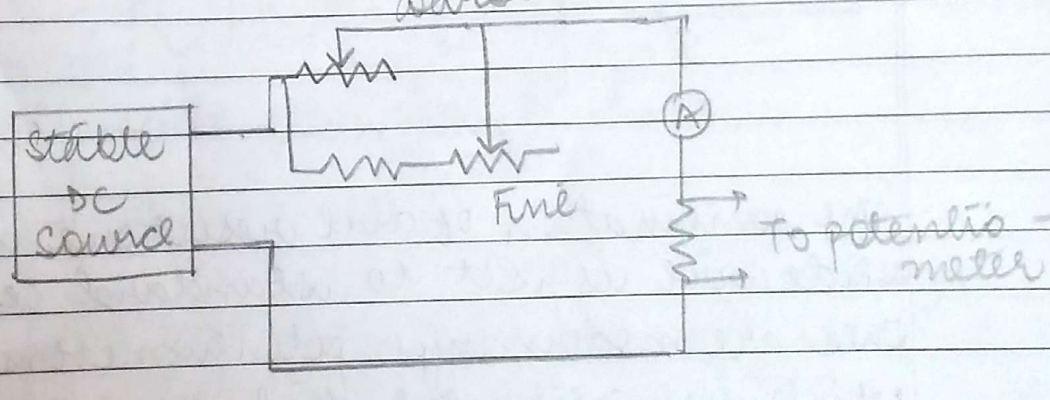
* measurement of resistance with potentiometer
→ DC supply should be stable
→ instruments to be calibrated must be DC
→ Potential divider is used for control of calibrated voltage



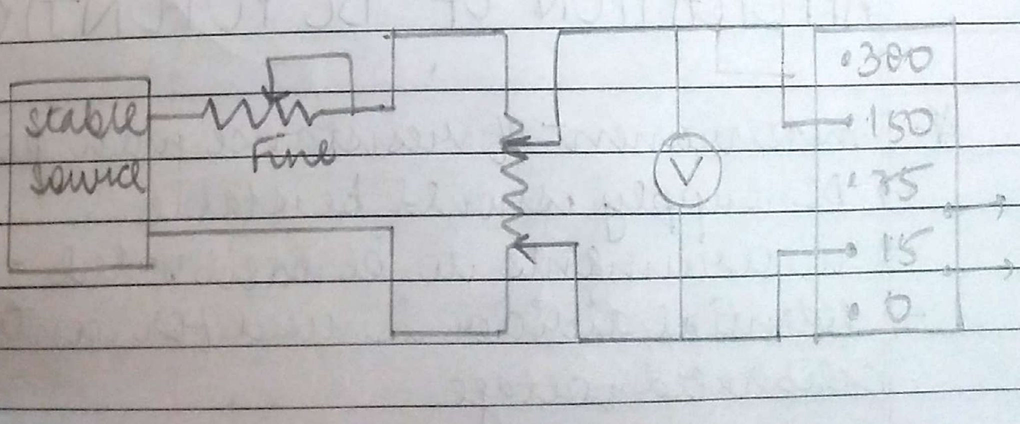
At 11'
Reading is VR
at 22'
Reading is Vs

$$\frac{V_S}{S} = \frac{V_R}{R}$$

* calibration of ammeter.

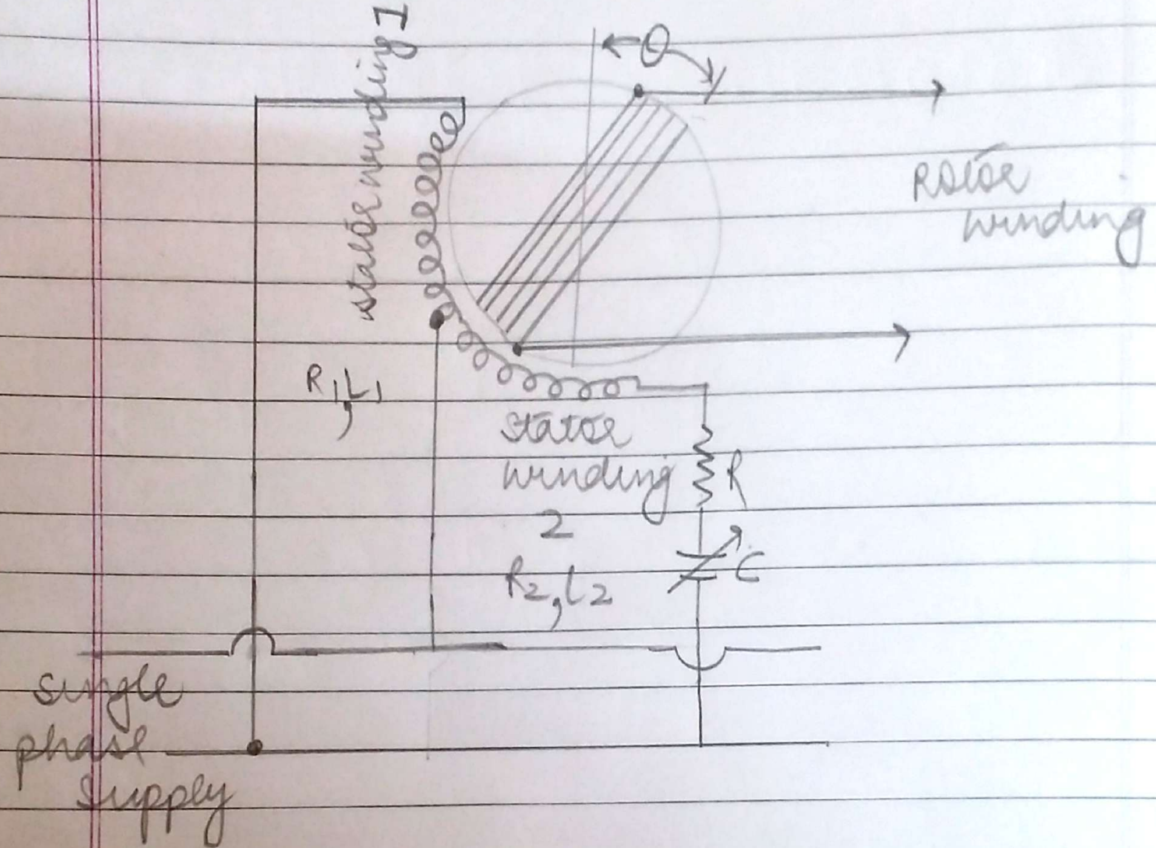


* Calibration of voltmeter



~~Drysdale~~

DRYSDALE PHASE SHIFTING TRANSFORMER



* Rotor can be adjusted to any required angle
* Phase displacement of rotor emf = angle through which rotor has been moved from its 0 position.
* Scale & pointer indicates the angle.